



TRƯỜNG ĐẠI HỌC FPT

SOFTWARE DESIGN SPECIFICATION

Project Name (Code)

Table of Contents

I. Record of Changes.....	3
II. Software Design Document.....	4
1. High Level Design.....	4
1.1 Software Architecture.....	4
1.2 Package Diagram.....	4
1.3 Database Design.....	5
2. State Transition Diagrams.....	7
2.1 PIN Validation.....	7
2.2	7
3. Detailed Design.....	8
3.1 <Feature/Function Name1>.....	8
3.2 <Feature/Function Name2>.....	9

I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

II. Software Design Document

1. High Level Design

1.1 Software Architecture - Ngọc

[The content of this section includes the overall architectural diagram which includes the sub-systems and/or components, the external systems (if any), and the relationships (communication messages) among them. You need also provide the explanation for each of the diagram components (modules, sub-systems, external systems, etc.).]

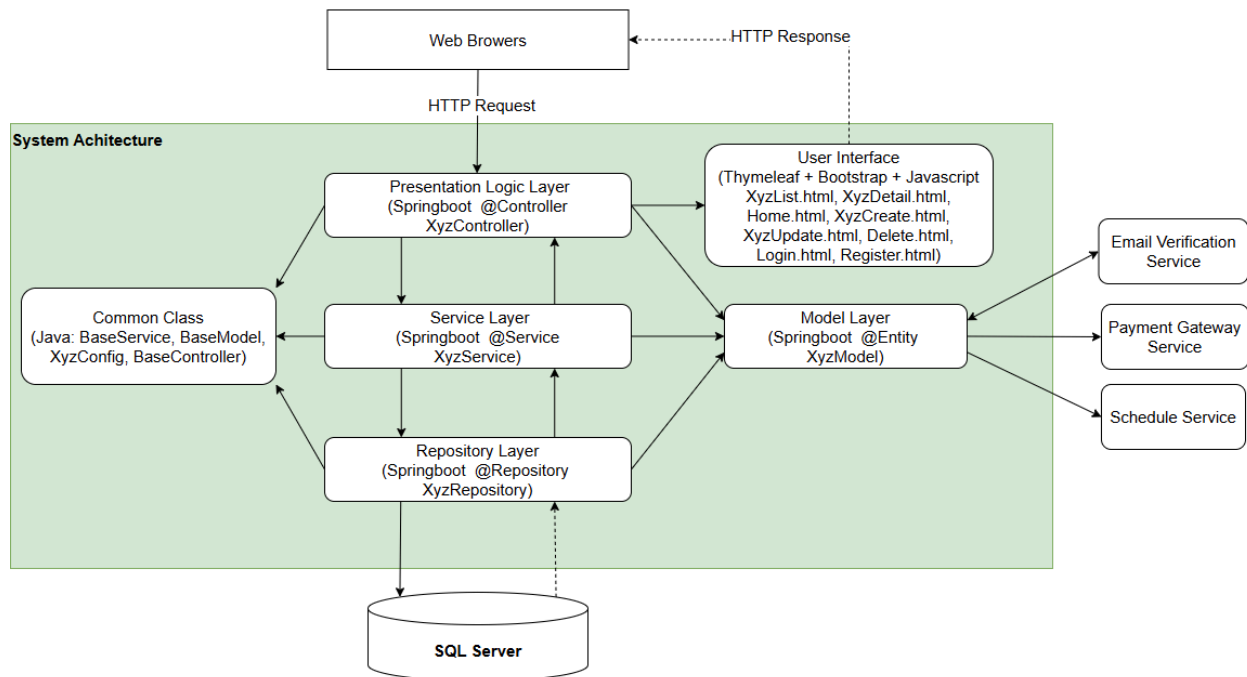
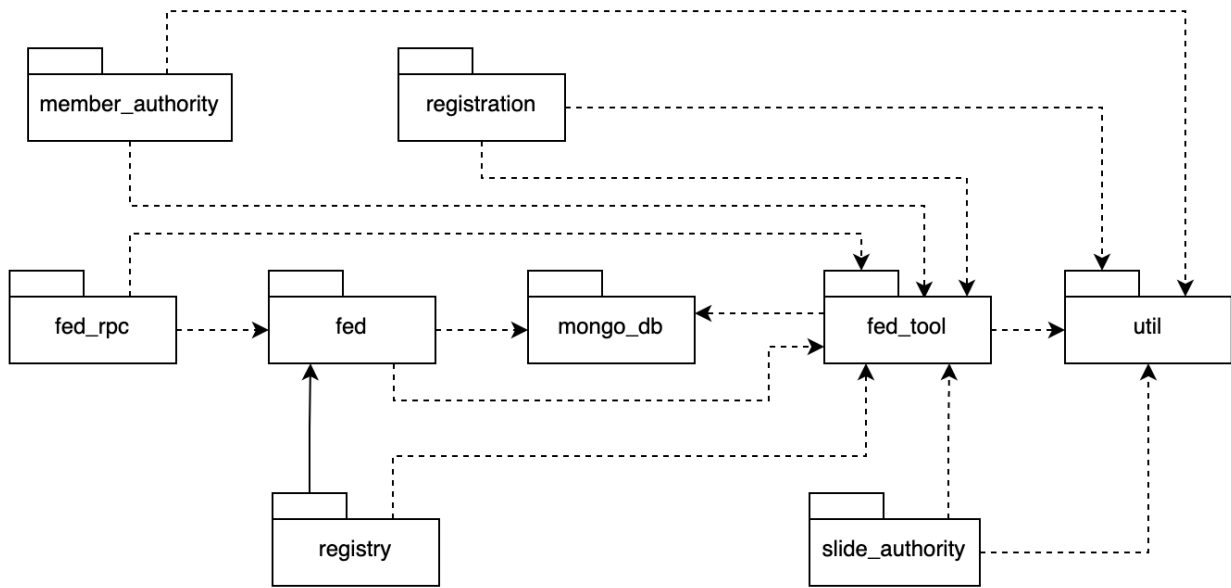


Figure X - Software Architecture

1.2 Package Diagram - Hải

[Provide the package diagram for each sub-system. The content of this section including the overall package diagram, the explanation, package and class naming conventions in each package. Please see the sample & description table format below]



Package descriptions

No	Package	Description
01	<i>Member_authority</i>	<i><Description of the package></i>
02	<i>registration</i>	<i><Description of the package></i>
03	...	

1.3 Database Design - Ngọc

[Provide the files description, database table relationship & table descriptions like example below]

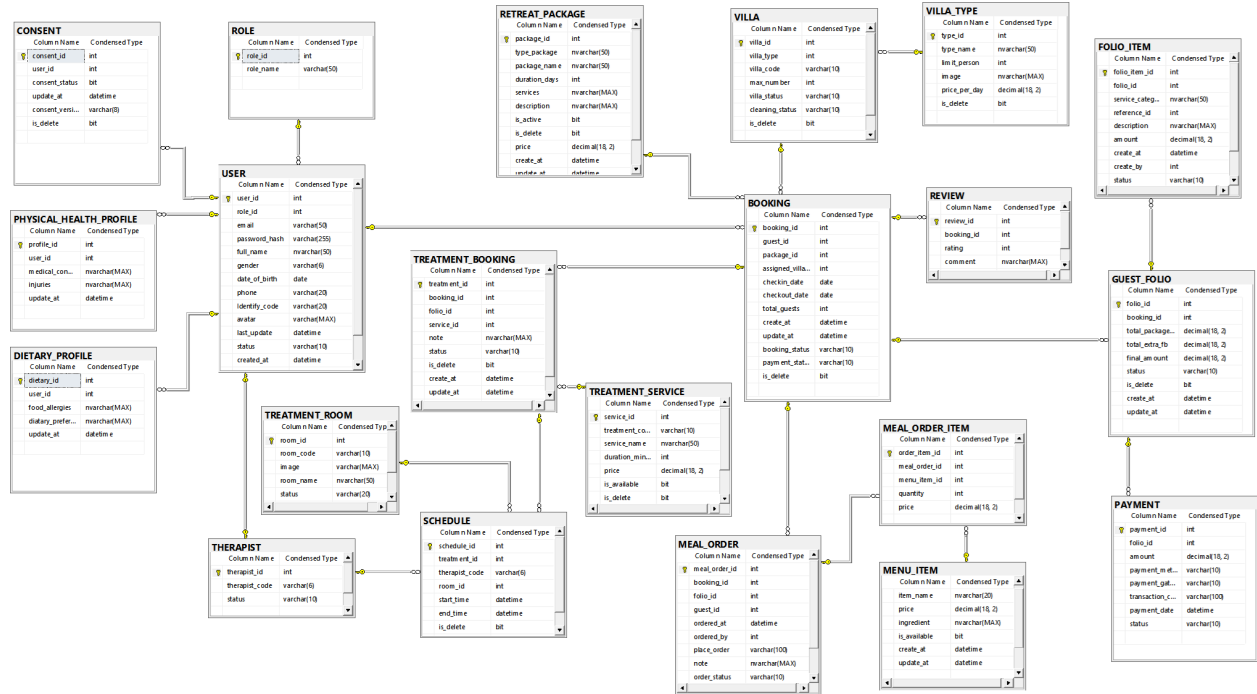


Figure X - Database Design Diagram

1.3.1 ROLE

No	Field	PK	FK	UN	NN	Description
1	role_id	X			X	Identify code (auto-increment)
2	role_name				X	Names of the roles in the system (Guest, Receptionist, Therapist, Chef, Admin)

1.3.2 USER

No	Field	PK	FK	UN	NN	Description
1	user_id	X			X	Unique User Identifier (Auto-incrementing)
2	role_id		X		X	Code associated with the ROLE table to identify data access rights according to the RBAC model
3	email				X	Unique email address used to log in and authenticate the SSO account
4	password_hash				X	User account password (securely hashed)
5	full_name					Full name of the user or customer.
6	gender					User's gender (Male / Female / Other).
7	date_of_birth					User's date of birth.
8	phone					User's contact phone number.
9	Identify_code					Personal identification number (Citizen Identification Card/Passport).Encryption at rest is

						mandatory for residence declaration purposes according to the Residence Law 2020
10	avatar					User's profile picture link.
11	last_update					Last account information update time (Default current time).
12	status					User account status (Active / Inactive).
13	created_at					Account creation time in the system.
14	last_login					User's last login time to the system.

1.3.3 THERAPIST

No	Field	PK	FK	UN	NN	Description
1	therapist_id	X	X		X	Specialist identifier, also a foreign key associated with USER(user_id)
2	therapist_code				X	Unique identifier code for each specialist
3	status					Current work status of the specialist (Available / Busy / Leave).

1.3.4 CONSENT

No	Field	PK	FK	UN	NN	Description
1	consent_id	X			X	Unique identifier of the consent history record (Auto-incrementing).
2	user_id		X		X	Code associated with the customer account USER(user_id)
3	consent_status					Customer consent status (1: Accepted, 0: Not accepted)
4	update_at					Time of the most recent consent status update.
5	consent_version					Version of the sensitive personal data consent terms.
6	is_delete					Mark up consent is deleted

1.3.5 DIETARY_PROFILE

No	Field	PK	FK	UN	NN	Description
1	dietary_id	X			X	Unique identifier of the nutrition profile (Auto-incrementing).
2	user_id		X		X	Customer account link code USER(user_id), to manage the private record
3	food_allergies					Food allergy information (e.g., peanut allergy, seafood allergy)
4	dietary_preference					Customer's personal dietary preferences (e.g., vegan)
5	update_at					Last updated nutrition profile date.

1.3.6 PHYSICAL_HEALTH_PROFILE

No	Field	P K	F K	UN	NN	Description
1	profile_id	X			X	Unique identifier of the physical health record (Auto-incrementing).
2	user_id		X		X	Code linked to the customer account USER(user_id) to manage the private record
3	medical_conditions					Underlying medical conditions, medical history (e.g., back pain, cardiovascular).
4	injuries					Customer injury information. Spa therapists use this for pre-treatment checks
5	update_at					Last updated physical health record time.

1.3.7 TREATMENT_SERVICE

No	Field	PK	FK	UN	NN	Description
1	service_id	X			X	Unique identifier for the treatment service (Auto-incrementing).
2	treatment_code				X	Unique abbreviation for the treatment service (maximum 10 characters).
3	service_name					Display name of the treatment/Spa service (e.g., Swedish Massage).
4	duration_minutes					Service duration in minutes (e.g., 60 minutes, 90 minutes).
5	price					Listed price of the treatment service (VND/USD)
6	is_available					Service availability status (1: Available, 0: Temporarily suspended).
7	is_delete					Flag to remove service logic from the catalog management system

1.3.8 TREATMENT_ROOM

No	Field	PK	FK	UN	NN	Description
1	room_id	X			X	Unique identifier for the treatment room (Auto-incrementing).
2	room_code				X	Unique identification code for the treatment room
3	image					Path to a real-life photo of the treatment room's facilities.
4	room_name					Name or display number of the Spa treatment room.
5	status					Treatment room status (Available / Occupied / Maintenance).
6	is_delete					Flag to remove the treatment room logic from the system.

1.3.9 VILA_TYPE

No	Field	PK	FK	UN	NN	Description
1	type_id	X			X	Unique identifier for the Villa type (Auto-incrementing).
2	type_name					Villa category name (e.g., Ocean View Villa)
4	image					Path to save the perspective image of the Villa type.
5	price_per_day					Villa rental price per day
6	is_delete					Flag to remove the Villa type logic from the system management catalog

1.3.10 VILA

No	Field	PK	FK	UN	NN	Description
1	villa_id	X			X	The unique identifier of the physical Villa (Auto-incrementing).
2	villa_type		X		X	Foreign key linking to the room type category VILLA_TYPE(type_id)
3	villa_code				X	Unique villa room number used in operation (e.g., VIL-101)
4	max_number					Maximum number of guests that can actually be accommodated at this Villa.
5	villa_status					Current booking status of the Villa (Available / Occupied / Reserved)
6	cleaning_status					Room cleaning status for check-in/out (Cleaned / Dirty / Needs Cleaning)
7	is_delete					Flag to remove the Villa entity logic from the system.

1.3.11 RETREAT_PACKAGE

No	Field	PK	FK	UN	NN	Description
1	package_id	X			X	Unique identifier of the all-inclusive wellness retreat package (Automatically incrementing).
2	type_package					Treatment package goal classification
3	package_name					Name of the GWI standard wellness retreat package
4	duration_days					Total number of days of stay for the treatment package
5	services					Description of the list of included treatment services in this all-inclusive package.
6	description					Detailed description of the itinerary and health benefits of the package.
7	is_active					Activation status allowing customers to search and purchase the package (1: Yes)
8	is_delete					Flag to remove the treatment package logic from the system catalog.
9	price					Total package price including accommodation
10	create_at					Time of creation of the treatment package in the system.
11	update_at					Time of last package information update.

1.3.12 BOOKING

No	Field	PK	FK	UN	NN	Description
1	booking_id	X			X	Unique identifier of the resort reservation (Auto-incrementing).
2	guest_id		X		X	Foreign key associated with USER(user_id) representing the customer who made the reservation
3	package_id		X			Foreign key associated with the RETREAT_PACKAGE(package_id) the customer selected
4	assigned_villa_id		X			Foreign key associated with the actual VILLA(villa_id) assigned by the Reception during Check-In
5	checkin_date					Actual or expected check-in date.
6	checkout_date					Actual or expected check-out date.
7	total_guests					Total number of guests staying in this reservation.
8	create_at					Time the customer created the online reservation.
9	update_at					Time the reservation information was last updated.

10	booking_status					Processing status of the reservation (Pending / Confirmed / Checked-In / Checked-Out / Cancelled).
11	payment_status					Payment status for deposit or room rate (Unpaid / Deposited / Fully-Paid)
12	is_delete					Flag indicating cancellation or deletion of booking logic.

1.3.13 GUEST_FOLIO

No	Field	PK	FK	UN	NN	Description
1	folio_id	X			X	Unique identifier of the room debt record (Auto-incrementing).
2	booking_id		X		X	Main foreign key linked to BOOKING (booking_id) to consolidate all incurred costs into the room account
3	total_package_amount					Total cost of room and initial fixed package treatment.
4	total_extra_fb					Total of additional costs outside the package from à la carte dining services
5	final_amount					Total final cost to be paid before departure
6	status					Room debt record status (Pending / Settled). Constraint to prevent Check-out if there is outstanding debt
7	is_delete					Flag to mark the logical deletion of the debt record.
8	create_at					Time the debt account was opened
9	update_at					Time of the most recent financial data update.

1.3.14 TREATMENT_BOOKING

No	Field	PK	FK	UN	NN	Description
1	treatment_id	X			X	Unique identifier for the Spa treatment appointment (Auto-incrementing).
2	booking_id		X		X	Foreign key linked to the original booking BOOKING(booking_id) to verify the stay
3	folio_id		X			Foreign key linked to the debt record GUEST_FOLIO to charge for additional services purchased outside the package
4	service_id		X		X	Foreign key linked to the treatment service category TREATMENT_SERVICE(service_id)
5	note					Special notes from the client or instructions from the doctor/specialist for the treatment session.
6	status					Spa appointment status (Scheduled / Completed / No-Show / Cancelled)

7	is_delete					Flag to mark the logical deletion of the Spa appointment.
8	create_at					Time of appointment booking.
9	update_at					Time of the most recent update of the treatment schedule information.

1.3.15 SCHEDULE

No	Field	PK	FK	UN	NN	Description
1	schedule_id	X			X	Unique identifier for the detailed resource allocation schedule (Auto-incrementing).
2	treatment_id		X		X	Foreign key linked to the original treatment appointment schedule TREATMENT_BOOKING(treatment_id).
3	therapist_code		X			Foreign key linked to the therapist code THERAPIST(therapist_code).
4	room_id		X			Constraint to prevent double-booking, Foreign key linked to the Spa room TREATMENT_ROOM(room_id).
5	start_time					Actual Spa treatment start time
6	end_time					Estimated treatment end time based on service duration.
7	is_delete					Flag to clear resource allocation schedule logic.

1.3.16 MENU_ITEM

No	Field	PK	FK	UN	NN	Description
1	menu_item_id	X			X	Unique identifier for the dish/drink on the menu (Automatically incremented).
2	item_name					Display name of the dish or drink on the menu
3	price					Retail price of the dish served as an add-on outside of the nutritional package
4	ingredient					Detailed ingredient list of the dish.Used for the system to automatically check customer allergy filters
5	is_available					Status of the dish available in the kitchen on that day (1: Available 0: Out of stock).
6	create_at					The time dish was created on the F&B management system.
7	update_at					Time of the last update of the dish's menu ingredient information.

1.3.17 MEAL_ORDER

No	Field	PK	FK	UN	NN	Description
----	-------	----	----	----	----	-------------

1	meal_order_id	X			X	Unique identifier of the food order (Auto-incrementing).
2	booking_id		X		X	Foreign key associated with the BOOKING(booking_id) resort booking to identify the room receiving the food
3	folio_id					Foreign key associated with the room payment record for accounting if additional costs are incurred
4	guest_id					Identifier of the customer making the food request or enjoying the food service.
5	ordered_at					Time of food order placement (Default current time).
6	ordered_by					User ID of the person creating the order (Customer places the order themselves or F&B staff places it on their behalf).
7	place_order					Delivery location
8	note					Customer's special dish note
9	order_status					Processing status of the meal order updated by the Chef (Preparing -> Ready for Delivery)

1.3.18 MEAL_ORDER_ITEM

No	Field	PK	FK	UN	NN	Description
1	order_item_id	X			X	Unique identifier for the item details line in the order (Auto-incrementing).
2	meal_order_id		X		X	Foreign key directly linked to the overall food order invoice MEAL ORDER(meal_order_id).
3	menu_item_id		X		X	Foreign key linked to the specific item called in MENU ITEM(menu_item_id).
4	quantity					Quantity of servings of the specific item requested in the order.
5	price					Unit price applicable at the time of placing the order (Used to fix the financial price).

1.3.19 FOLIO_ITEM

No	Field	PK	FK	UN	NN	Description
1	folio_item_id	X			X	Unique identifier for the service charge line (Automatically increments).
2	folio_id		X		X	Foreign key directly linked to the customer's aggregated debt account GUEST FOLIO(folio_id)
3	service_category					Classification of the generated service group for revenue analysis reporting (e.g., Extra Spa, Extra F&B)
4	reference_id					Reference ID of the generated original order
5	description					Detailed explanation of the incurred debt item for customer verification upon check-out

6	amount					Value of the generated service charge (Will be automatically added to the aggregated debt Folio)
7	create_at					Time of this individual charge transaction to be added to the room account.
8	create_by					Identifier of the employee who pushed this charge to the guest's room system.
9	status					Payment processing status for additional charges

1.3.20 REVIEW

No	Field	PK	FK	UN	NN	Description
1	review_id	X			X	Unique identifier of the feedback record (Auto-incrementing).
2	booking_id		X		X	Foreign key associated with the BOOKING(booking_id) resort reservation after check-out
3	rating					Quantitative service quality rating (Constraint value from 1 to 5 stars)
4	comment					Content of the customer's free-form text feedback about the resort

1.3.21 PAYMENT

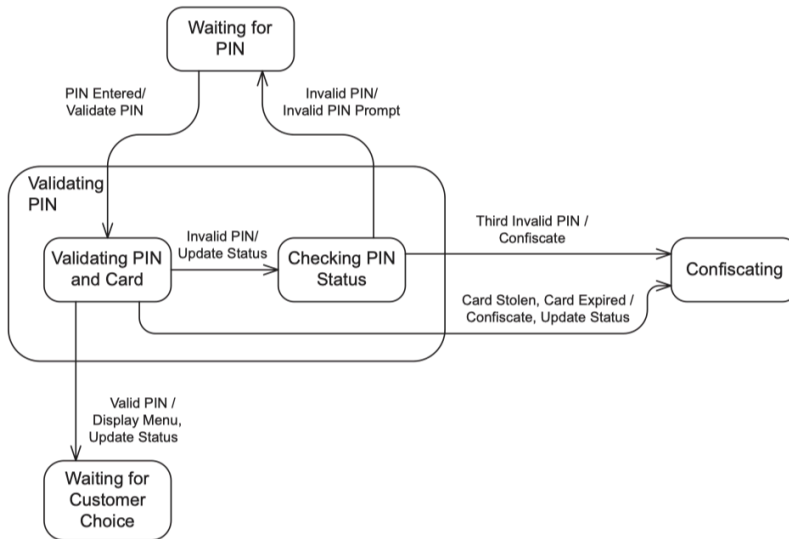
No	Field	PK	FK	UN	NN	Description
1	payment_id	X			X	Unique identifier for the financial payment transaction record (Auto-incrementing).
2	folio_id		X			Foreign key directly linked to the aggregated debt account record GUEST FOLIO(folio_id)
3	amount					Actual amount of the payment transaction
4	payment_method					Applicable payment method
5	payment_gateway					Name of the third-party online payment gateway API processing the transaction (Stripe / VNPay / PayPal)
6	transaction_code				X	Transaction tracking code returned from the bank system or partner payment gateway for reconciliation
7	payment_date					Time the payment transaction was successfully recorded
8	status					Status of the financial payment transaction processing (Success / Failed / Processing).

2. State Transition Diagrams - Hải

[Specify and draw state charts (state transition diagrams) for the data and system like below sample. In the diagrams, beside the states, you are required to provide suitable events, actions on the state transitions, entry actions, or exit actions]

2.1 PIN Validation

[Provide state chart with extra explanations if needed]

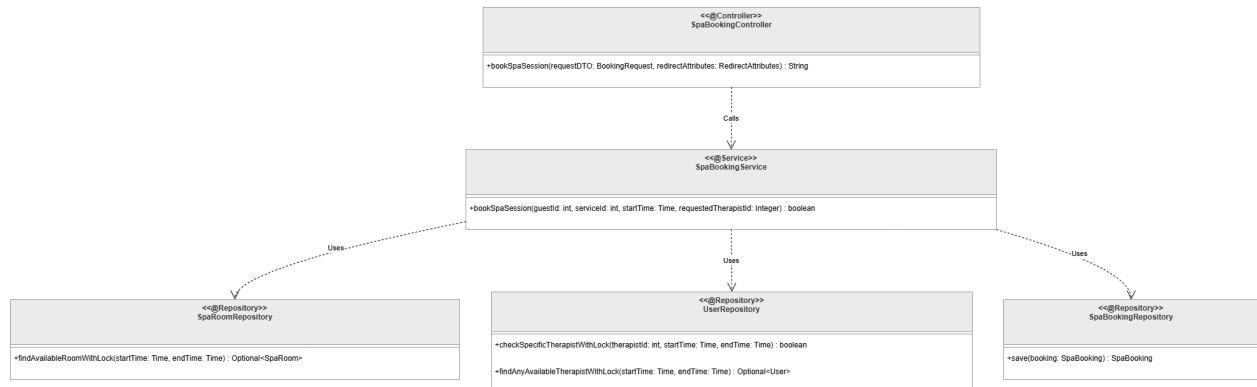


2.2 ...

3. Detailed Design - Dương

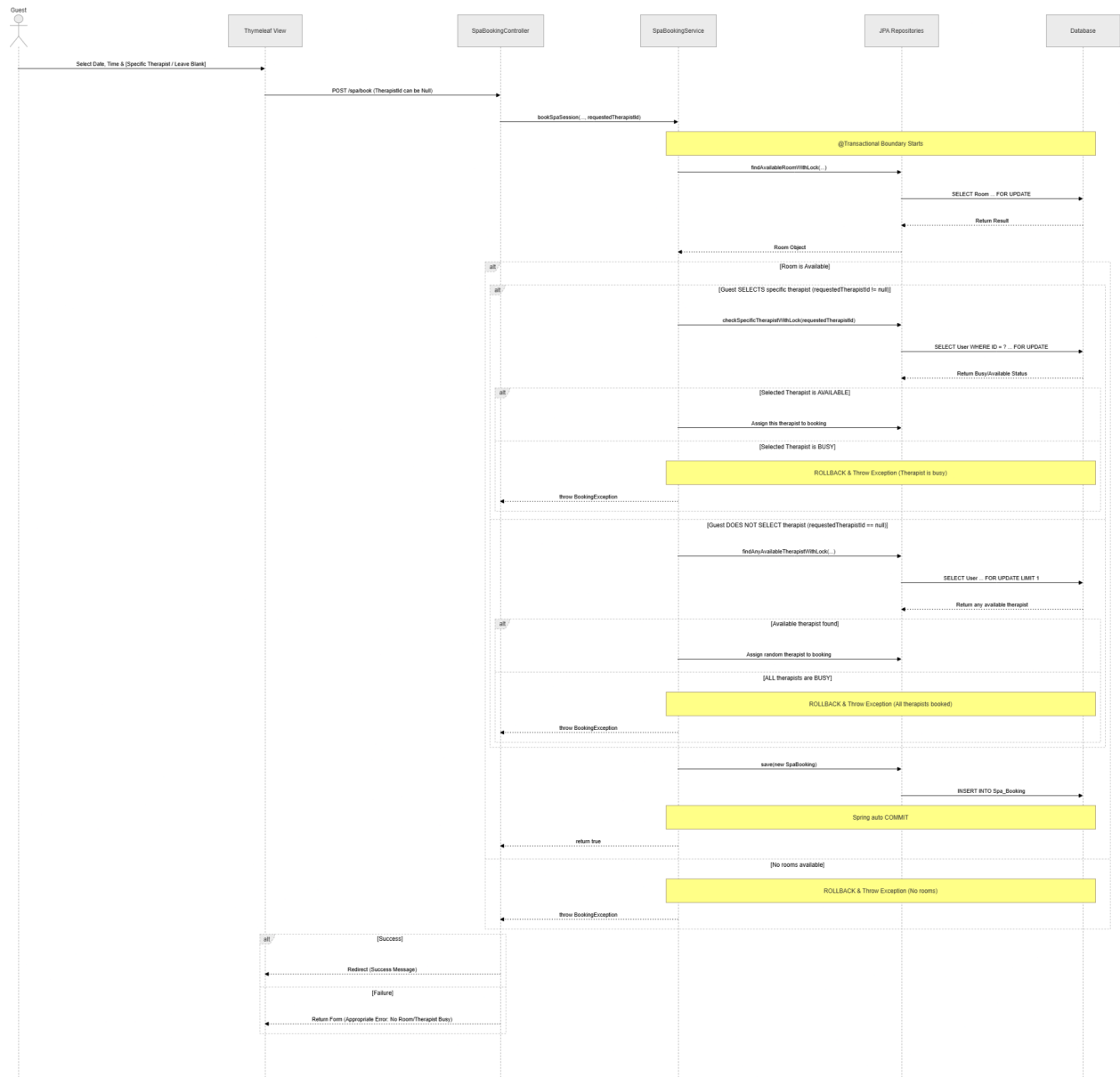
3.1 Auto-Matching Spa Scheduling & Specific Therapist Selection (UC12) This feature allows guests to book a Spa session with two options: assigning a specific Therapist, or letting the system automatically assign any available therapist. The system ensures data locking (**FOR UPDATE**) and automatic Rollback via **@Transactional** to prevent room and personnel resource collisions (double-booking).

3.1.1 Class Diagram The main difference here is that the **requestedTherapistId** parameter is passed to the Service as a reference type (Nullable) to identify whether the guest has specified a therapist or not.



[Link](#)

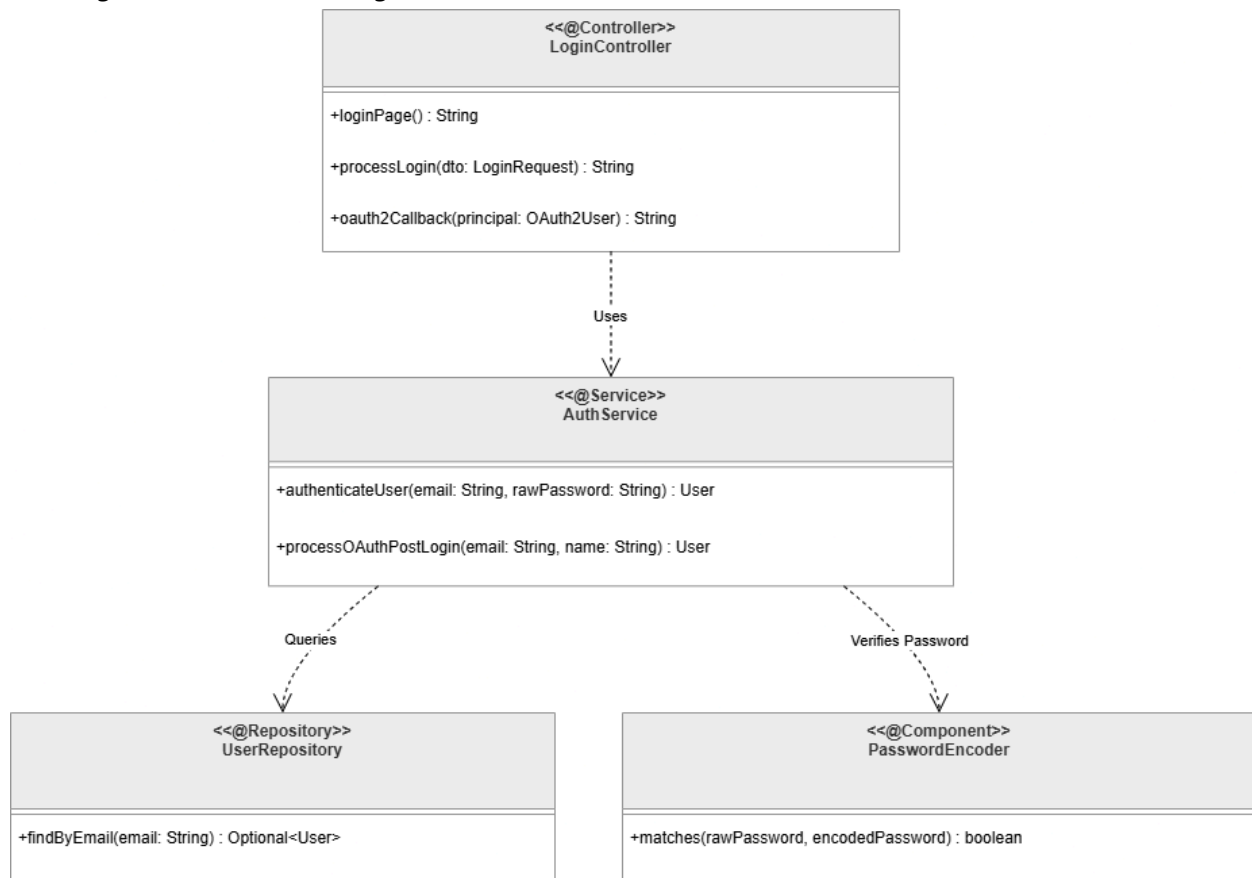
3.1.2 Sequence Diagram: Spa Booking Branch Processing This sequence diagram includes an additional condition block (Alt) to branch the therapist search algorithm based on the guest's selection.



[Link](#)

3.2 Login & Authentication (UC01) The system supports user authentication via two methods: traditional login (Email/Password) utilizing BCrypt encryption, and single sign-on via Google OAuth2. Upon successful authentication, the system leverages Spring Security to manage user sessions and assign corresponding roles (Guest, Admin, Therapist, etc.).

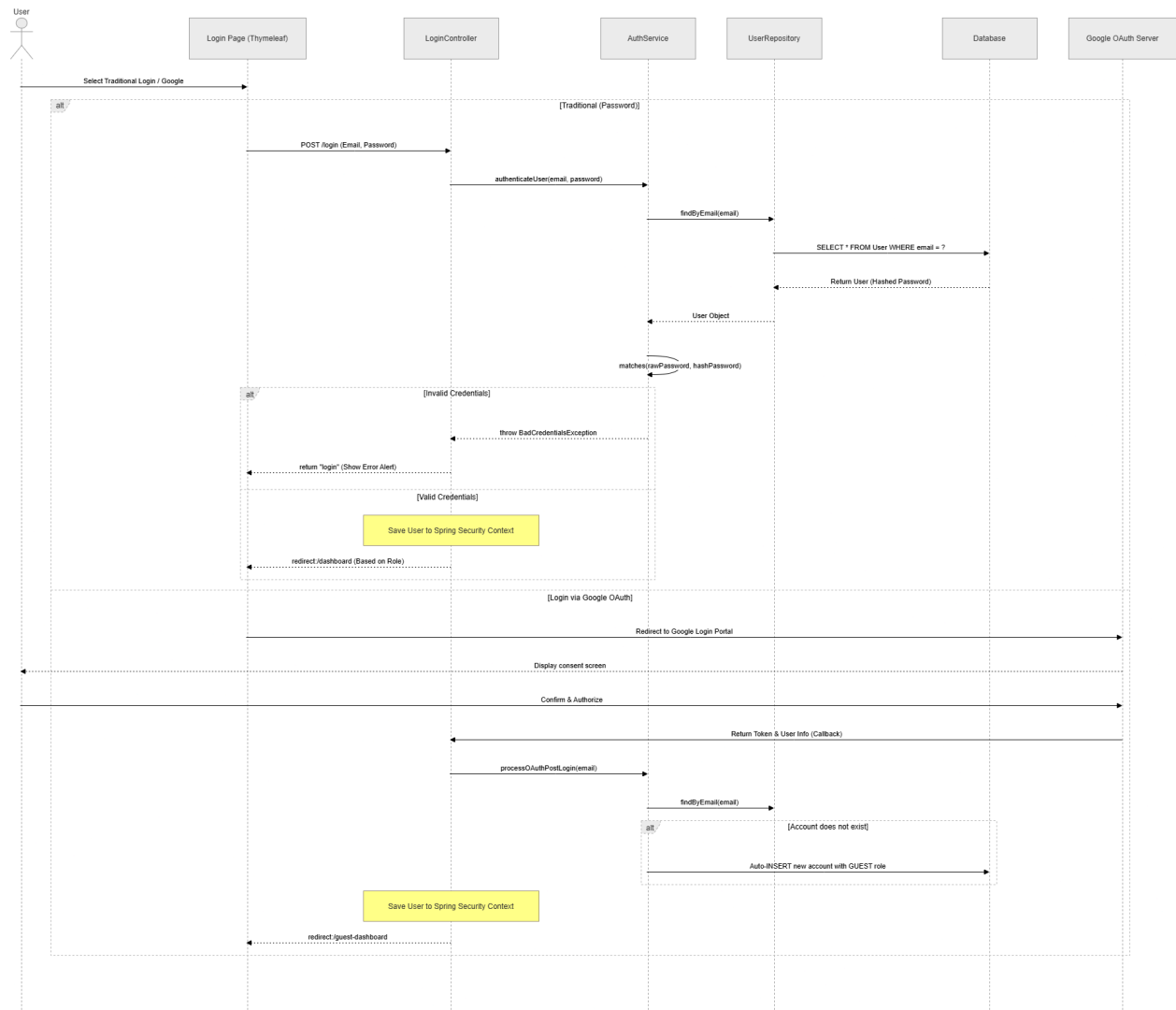
3.2.1 Class Diagram This diagram illustrates the Spring Security components and the Service layer handling the authentication logic.



[Link](#)

3.2.2 Sequence Diagram: Authentication Processing

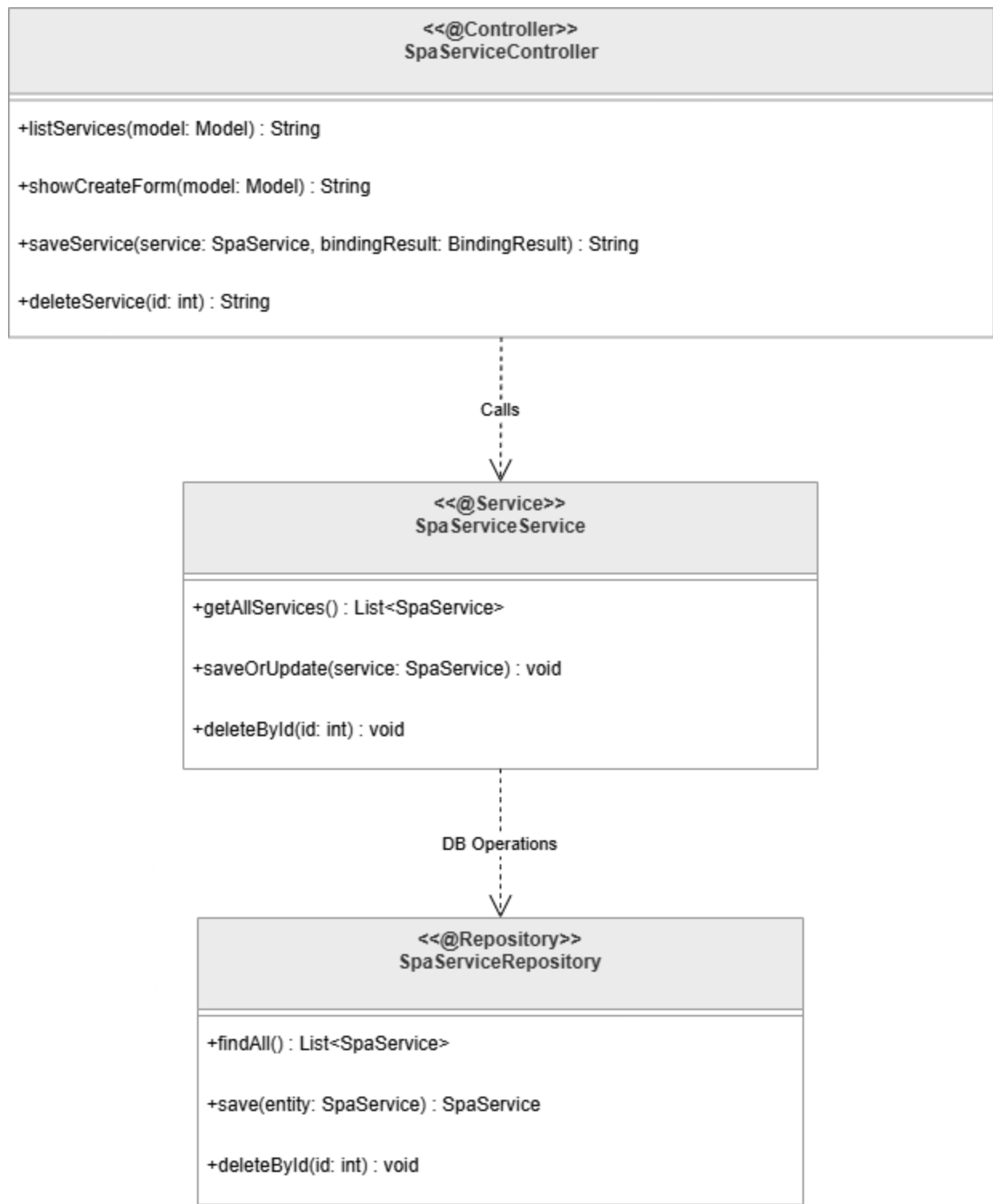
The sequence diagram below covers both the traditional login flow and the branching logic for the Google OAuth API integration.



[Link](#)

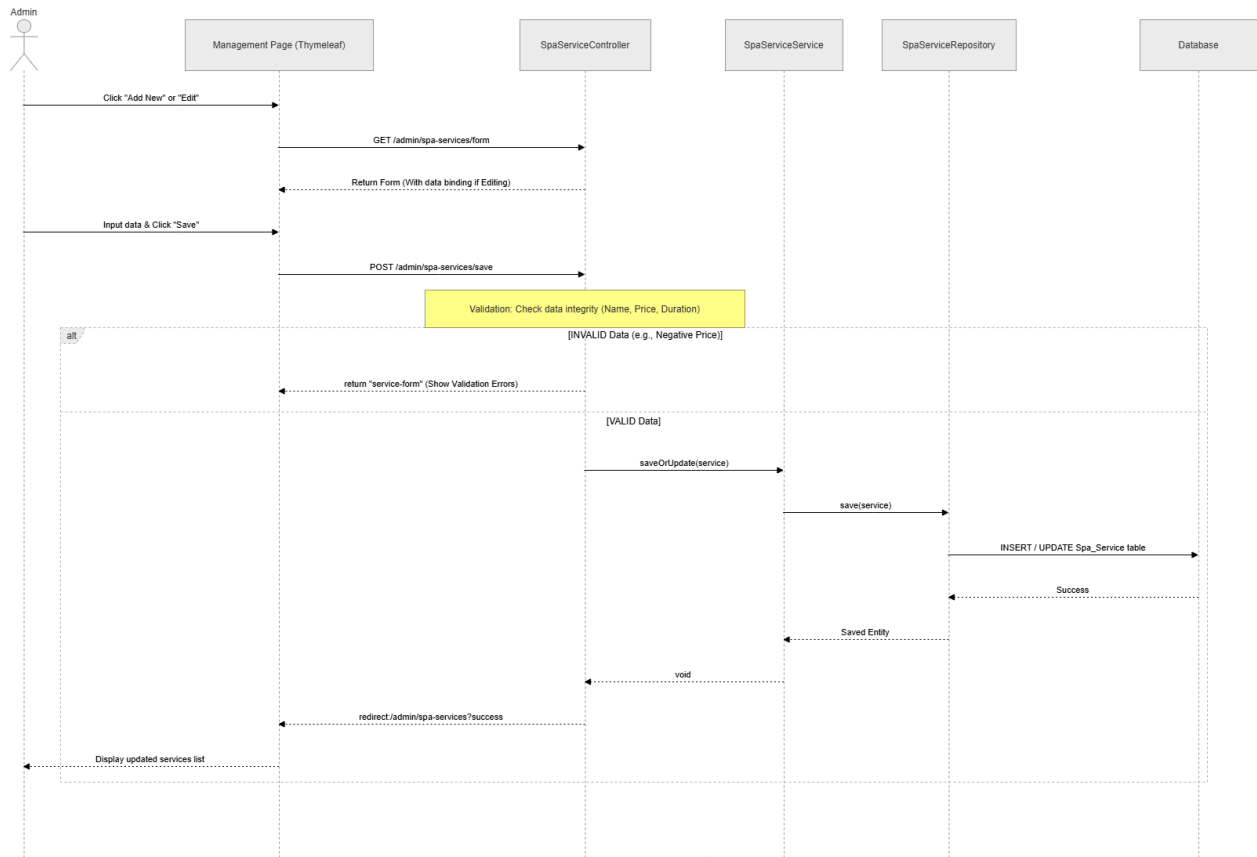
3.3 Manage Spa Services (Standard CRUD Template) This feature allows the Admin to manage the master data for Spa services (Create, Read, Update, Delete). The update process uses Server-Side Rendering to bind data to forms and handles persistence via Spring Data JPA. *(Note: This class structure and data flow serve as the standard reference architecture for all other master data management features, such as Manage Staff, Manage Rooms, etc.).*

3.3.1 Class Diagram



[Link](#)

3.3.2 Sequence Diagram: Create/Update Service This diagram illustrates the flow for adding a new Spa service or updating an existing one. The Read and Delete flows follow a structurally identical execution path.



[Link](#)